

Descriptive Statistics

Mathematics for Computing – Module 3

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September 2026

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1. Statistics

Definition

Statistics is the science of collecting, organizing, summarizing, analyzing, interpreting, and presenting data to obtain meaningful information and support decision-making.

Main branches

- **Descriptive Statistics** – summarizes and presents data.
- **Inferential Statistics** – uses sample data to draw conclusions about a population.

Example

For the marks of 50 students, calculating the mean and standard deviation describes the data, while using the sample to estimate the average marks of the entire class is an inferential task.

2. Data

Definition

Data are raw facts, observations, measurements, or values collected for analysis.

Examples

- Student marks: 72, 85, 64, 91
- Age: 20, 21, 22, 20
- Temperature: 28.5°C
- Image pixels represented by numerical values

Mathematical Representation

A dataset can be represented as

$$X = \{x_1, x_2, \dots, x_n\}$$

where x_i represents an observation and n is the number of observations.

Application: Data forms the basic input for statistical analysis and machine learning models.

3. Information

Definition

Information is processed, organized, and interpreted data that provides meaning.

Example

Suppose the marks are

60, 70, 80, 90, 100

The raw values are **data**.

Their average,

$$\bar{x} = 80,$$

provides meaningful information about the overall performance.

Key idea

Data → Processing → Information

4. Knowledge

Definition

Knowledge is the understanding obtained from information by identifying relationships, patterns, rules, or meaningful insights.

Example

If analysis shows that students who regularly attend classes generally obtain higher marks, this observed relationship contributes to knowledge.

Flow

Data → Information → Knowledge

Application

In computing and AI, knowledge can be used to build rules, models, and systems that support prediction and decision-making.

5. Intelligence

Definition

Intelligence is the ability to use knowledge to learn, reason, solve problems, make decisions, and adapt to new situations.

Example

A machine learning system can:

- 1 Learn patterns from data.
- 2 Convert patterns into useful information.
- 3 Build knowledge from these patterns.
- 4 Use the learned knowledge to make predictions or decisions.

Conceptual Flow

Data → Information → Knowledge → Intelligence

6. Types of Data

Data can be classified according to the number of variables being observed.

Univariate Data

Data involving **one variable**.

Bivariate Data

Data involving **two variables**, often used to study their relationship.

Multivariate Data

Data involving **three or more variables**.

6.1 Univariate Data

Definition

Univariate data contains observations of a single variable.

Example

Heights of five students:

$$X = \{160, 165, 170, 172, 168\}$$

Here, the only variable is **height**.

Common analysis

- Mean
- Median
- Mode
- Variance
- Standard deviation

Application: Used to understand the distribution and variability of a single feature.

6.2 Bivariate Data

Definition

Bivariate data contains observations of two variables measured on the same subjects.

Example

Consider years of education and income:

$$X = \text{Years of Education}, \quad Y = \text{Income}$$

Each observation is represented as a pair

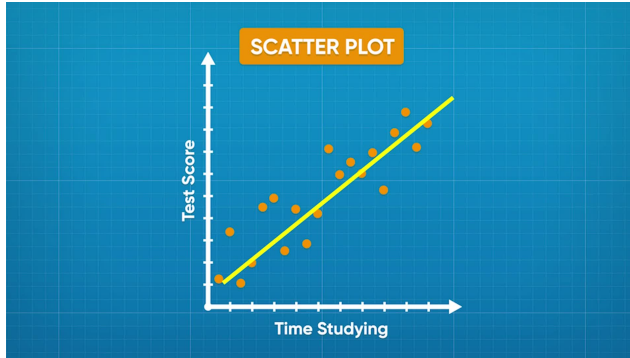
$$(x_i, y_i).$$

Common analysis

- Covariance
- Correlation
- Simple linear regression

Application: Used to study relationships between two variables.

6.2 Bivariate Data – Figure



The figure represents bivariate data by showing the relationship between two variables.

Each point represents one observation with a value for both variables.

Key Observation

The pattern of the points helps us identify whether the two variables have a possible relationship.

6.3 Multivariate Data

Definition

Multivariate data contains observations involving three or more variables.

Example A student dataset may contain:

$$\{\text{Age, Attendance, Study Hours, Marks}\}$$

Each observation can be represented as

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})$$

where p is the number of variables.

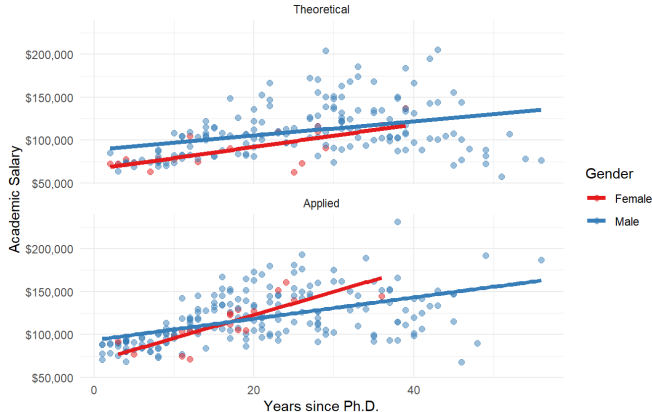
Common analysis

- Covariance matrix
- Correlation analysis
- Multiple regression

Application: Most real-world AI and machine learning datasets are multivariate.

6.3 Multivariate Data – Figure

Relationship of salary and years since degree by sex and discipline
9-month salary for 2008-2009



The figure represents data involving more than two variables simultaneously. Each observation contains multiple measurements.

Observation

The figure shows how several variables can be examined together to identify patterns and relationships in the data.

7. Measures of Central Tendency

Definition

Measures of central tendency describe the **central or typical value** of a dataset.

The three commonly used measures are:

- **Mean** – arithmetic average of the observations.
- **Median** – middle value after arranging the data.
- **Mode** – most frequently occurring value.

Purpose

They provide a single value that represents the center of a dataset.

Application:

Used to summarize datasets in statistics, data analysis, and machine learning.

7.1 Mean

Definition

The arithmetic mean is obtained by dividing the sum of all observations by the number of observations.

Formula

For observations $x_1, x_2, \dots, x_n$,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

where n is the number of observations.

Interpretation

The mean represents the value obtained when the total of all observations is distributed equally among the n observations.

Note: The mean uses every observation in the dataset and is therefore sensitive to extreme values.

7.1 Mean – Example

Example

Consider the dataset:

$$X = \{10, 20, 30, 40, 50\}$$

There are $n = 5$ observations.

Calculation

$$\bar{x} = \frac{10 + 20 + 30 + 40 + 50}{5}$$

$$\boxed{\bar{x} = 30}$$

Interpretation

The average value of the dataset is 30.

Application

The mean is widely used to summarize numerical features such as marks, income, temperature, and sensor measurements.

7.2 Median

Definition

The median is the **middle value** of an ordered dataset.

The observations must first be arranged in ascending or descending order.

Formula

For an ordered dataset with n observations:

$$\text{Median} = \begin{cases} x_{\frac{n+1}{2}}, & n \text{ is odd} \\ \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}, & n \text{ is even} \end{cases}$$

Key idea

- If n is odd, the median is the single middle observation.
- If n is even, the median is the average of the two middle observations.

7.2 Median – Examples

Example 1 – Odd number of observations

Consider:

$$X = \{10, 20, 30, 40, 50\}$$

Here, $n = 5$ is odd.

$$\text{Median} = x_{\frac{5+1}{2}} = x_3 = \boxed{30}$$

Example 2 – Even number of observations

Consider:

$$X = \{10, 20, 30, 40\}$$

Here, $n = 4$ is even.

$$\text{Median} = \frac{x_2 + x_3}{2} = \frac{20 + 30}{2} = \boxed{25}$$

7.3 Median and Outliers

Example

Consider:

$$X = \{10, 20, 30, 40, 1000\}$$

The mean is

$$\bar{x} = \frac{10 + 20 + 30 + 40 + 1000}{5} = 220$$

while the median is

$$\text{Median} = 30$$

Interpretation

The extreme value 1000 greatly increases the mean, while the median remains close to the center of the majority of observations.

Key Point

- Mean is sensitive to extreme values.
- Median is more resistant to outliers.

7.4 Mode

Definition

The mode is the value that occurs with the **highest frequency** in a dataset.

Example

Consider:

$$X = \{2, 4, 4, 5, 6, 4, 7, 8\}$$

Therefore,

$\text{Mode} = 4$

Types

- **Unimodal** – one mode.
- **Bimodal** – two modes.
- **Multimodal** – more than two modes.

Application:

Useful for identifying the most common category or value, especially for categorical data.

7.5 Mean, Median and Mode

Measure	Main Idea	Sensitivity to Outliers
Mean	Arithmetic average	High
Median	Middle value	Low
Mode	Most frequent value	Generally low

When to use?

- **Mean:** Suitable when the data is approximately symmetric and does not contain strong outliers.
- **Median:** Suitable for skewed data or data containing extreme values.
- **Mode:** Useful when the most frequent value or category is important.

Application:

Choosing an appropriate measure of central tendency helps summarize real-world datasets effectively.

8. Measures of Dispersion

Definition

Measures of dispersion describe the **spread or variability** of observations around a central value.

Two datasets may have the same mean but different levels of variability.

Common measures

- Range
- Variance
- Standard deviation
- Interquartile range (IQR)

Purpose

They help determine how widely the observations are distributed and how consistent the data is.

8.1 Range

Definition

The range is the difference between the **maximum** and **minimum** values in a dataset.

Formula

$$\text{Range} = x_{\max} - x_{\min}$$

Key idea

- Uses only the smallest and largest observations.
- Provides a simple measure of the total spread.
- Highly sensitive to extreme values.

Application:

Useful for obtaining a quick estimate of the spread of numerical data.

8.1 Range – Example

Example

Consider the dataset:

$$X = \{10, 20, 30, 40, 50\}$$

The minimum value is

$$x_{\min} = 10$$

and the maximum value is

$$x_{\max} = 50$$

Therefore,

$$\text{Range} = 50 - 10 = \boxed{40}$$

Interpretation

The observations span a total interval of 40 units.

8.2 Variance

Definition

Variance measures the average squared deviation of observations from their mean.

For a dataset $x_1, x_2, \dots, x_n$ with mean $\bar{x}$,

Population variance:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Sample variance:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Key idea

The deviations are squared so that positive and negative deviations do not cancel each other.

8.2 Variance – Example

Example

Consider:

$$X = \{2, 4, 6\}$$

First calculate the mean:

$$\bar{x} = \frac{2 + 4 + 6}{3} = 4$$

Squared deviations:

$$(2 - 4)^2 = 4, \quad (4 - 4)^2 = 0, \quad (6 - 4)^2 = 4$$

Population variance:

$$\sigma^2 = \frac{4 + 0 + 4}{3} = \boxed{\frac{8}{3}}$$

Interpretation

A larger variance indicates greater spread around the mean.

8.3 Standard Deviation

Definition

Standard deviation is the positive square root of the variance.

Population standard deviation

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

Sample standard deviation

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Standard deviation is expressed in the **same units as the original data**, unlike variance.

8.3 Standard Deviation – Example

Example

For the dataset

$$X = \{2, 4, 6\},$$

$$\sigma^2 = \frac{8}{3}$$

Therefore,

$$\sigma = \sqrt{\frac{8}{3}} \approx \boxed{1.63}$$

Interpretation

The standard deviation indicates the typical magnitude of deviation of observations from the mean.

Application

Standard deviation is widely used in data analysis and machine learning to measure feature variability.

8.4 Interquartile Range (IQR)

Definition

The interquartile range measures the spread of the **middle 50%** of the observations.

The data is divided into four parts using quartiles:

- Q_1 – First quartile (25th percentile)
- Q_2 – Second quartile (50th percentile / median)
- Q_3 – Third quartile (75th percentile)

Formula

$$IQR = Q_3 - Q_1$$

Key idea

IQR focuses on the middle portion of the data and is less affected by extreme observations.

8.4 Interquartile Range – Example

Example

Consider the ordered dataset:

$$X = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

The median is

$$Q_2 = 5$$

Lower half

$$\{1, 2, 3, 4\}$$

Therefore,

$$Q_1 = \frac{2 + 3}{2} = \boxed{2.5}$$

8.4 Interquartile Range – Example

Upper half

$$\{6, 7, 8, 9\}$$

Therefore,

$$Q_3 = \frac{7 + 8}{2} = \boxed{7.5}$$

From the previous example:

$$Q_1 = 2.5, \quad Q_3 = 7.5$$

Quartiles

$$Q_1 = 2.5, \quad Q_2 = 5, \quad Q_3 = 7.5$$

8.4 Interquartile Range – Calculation

Formula

$$IQR = Q_3 - Q_1$$

Calculation

Using

$$Q_1 = 2.5, \quad Q_3 = 7.5$$

we get

$$IQR = 7.5 - 2.5$$

$$IQR = 5$$

Interpretation

The interquartile range is 5, meaning that the middle 50% of the observations are spread over an interval of 5 units.

Key Point

IQR is less affected by extreme values than the range.

8.5 Comparison of Dispersion Measures

Measure	What it describes	Outlier sensitivity
Range	Total spread	High
Variance	Squared deviation from mean	High
Standard deviation	Spread around mean	High
IQR	Spread of middle 50%	Low

When to use?

- **Range:** Quick measure of total spread.
- **Variance:** Useful in statistical and mathematical models.
- **Standard deviation:** Easy to interpret in the original units.
- **IQR:** Useful for skewed data and datasets with outliers.

9. Histogram

Definition

A histogram is a graphical representation of the distribution of numerical data using **continuous intervals called bins**.

Key features

- The horizontal axis represents the range of values.
- The vertical axis represents frequency or count.
- Adjacent bars touch because the intervals are continuous.

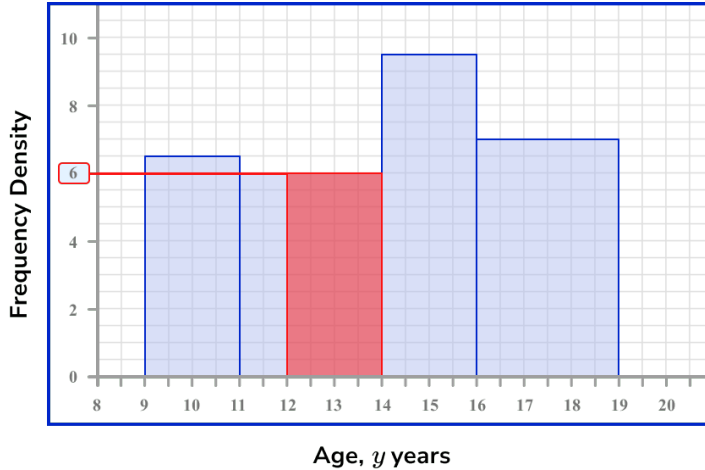
Purpose

A histogram helps identify the shape, center, spread, and possible unusual patterns in a dataset.

Application

Histograms are commonly used in exploratory data analysis to understand the distribution of numerical features.

9.1 Histogram – Figure



The observations are grouped into intervals called bins. The height of each bar represents the frequency of observations in that interval.

Observation

The overall shape of the bars provides a visual indication of the distribution of the data.

9.2 Histogram – Example

Example

Consider the marks of students:

$$X = \{42, 45, 48, 51, 52, 55, 57, 58, 61, 63, 65, 66, 68, 70, 72\}$$

The observations can be grouped into intervals such as:

Marks Interval	Frequency
40 – 49	3
50 – 59	5
60 – 69	4
70 – 79	3

The histogram represents these frequencies using adjacent bars for the intervals.

Interpretation

The distribution can be examined to understand where the observations are concentrated.

9.3 Histogram – Interpretation

A histogram can reveal important characteristics of a dataset.

- **Center:** Where most observations are concentrated.
- **Spread:** How widely the observations are distributed.
- **Shape:** Whether the distribution is symmetric, skewed, or has multiple peaks.
- **Possible outliers:** Observations that appear far from the main concentration.

Application in AI

Histograms can be used during exploratory data analysis to understand feature distributions before preprocessing or model building.

10. Box Plot

Definition

A box plot is a graphical representation of a dataset based on its **five-number summary**.

Five-number summary

- Minimum
- First quartile (Q_1)
- Median (Q_2)
- Third quartile (Q_3)
- Maximum

The box represents the interval from Q_1 to Q_3 .

$$IQR = Q_3 - Q_1$$

Purpose

A box plot summarizes the center, spread and potential outliers of numerical data.

10.1 Box Plot – Components

Main components

- **Lower whisker** – lower end of the non-outlier range.
- **Box** – extends from Q_1 to Q_3 .
- **Median line** – represents Q_2 .
- **Upper whisker** – upper end of the non-outlier range.
- **Points beyond whiskers** – potential outliers.

Outlier rule

An observation is commonly considered an outlier if

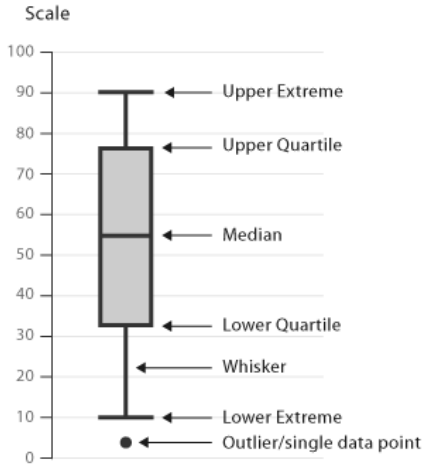
$$x < Q_1 - 1.5(IQR)$$

or

$$x > Q_3 + 1.5(IQR)$$

The box plot provides a compact summary of the distribution and is particularly useful for comparing multiple datasets.

10.2 Box Plot – Figure



The box represents the middle 50% of the data, from Q_1 to Q_3 . The line inside the box represents the median.

The whiskers extend toward the lower and upper values that are not considered potential outliers.

10.3 Box Plot – Example

Example

Consider the ordered dataset:

$$X = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

The quartiles are:

$$Q_1 = 2.5, \quad Q_2 = 5, \quad Q_3 = 7.5$$

Therefore,

$$IQR = Q_3 - Q_1 = 7.5 - 2.5 = 5$$

The lower and upper outlier boundaries are:

$$Q_1 - 1.5(IQR) = 2.5 - 7.5 = -5$$

$$Q_3 + 1.5(IQR) = 7.5 + 7.5 = 15$$

Since all observations lie between -5 and 15 , there are no outliers.

11. Outliers

Definition

An outlier is an observation that lies unusually far from the general pattern of the dataset.

Common causes

- Measurement or data-entry errors
- Unusual events
- Natural variation
- Rare observations

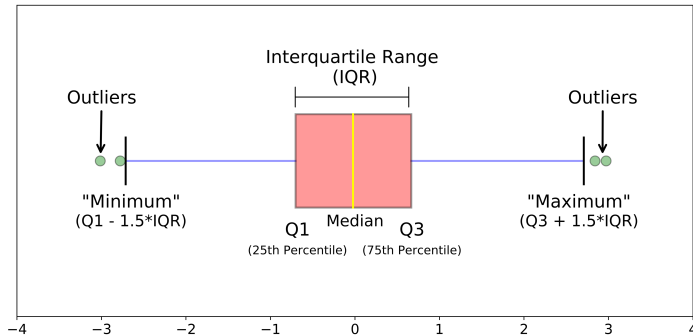
Effect

Outliers can strongly influence statistical measures such as the mean, variance, and standard deviation.

Important

An outlier is not necessarily an error. It should be investigated before deciding whether to remove or transform it.

11.1 Outliers – Figure



The figure shows an observation located far from the main group of observations.

Such observations may strongly influence statistics such as the mean and variance.

11.2 Outliers – IQR Method

The IQR method identifies potential outliers using the first and third quartiles.

Step 1: Calculate IQR

$$IQR = Q_3 - Q_1$$

Step 2: Calculate the boundaries

$$\text{Lower Bound} = Q_1 - 1.5(IQR)$$

$$\text{Upper Bound} = Q_3 + 1.5(IQR)$$

Step 3: Identify potential outliers

$$x < \text{Lower Bound}$$

or

$$x > \text{Upper Bound}$$

Observations outside these boundaries are treated as potential outliers.

12. Skewness

Definition

Skewness measures the **asymmetry** of a probability distribution or dataset around its mean.

Types

- **Symmetric:** Distribution is approximately balanced.
- **Right-skewed:** Longer tail extends toward larger values.
- **Left-skewed:** Longer tail extends toward smaller values.

Interpretation

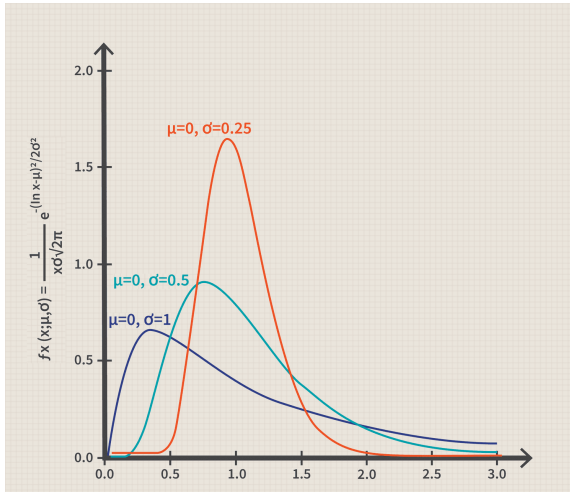
Right skewness $\Rightarrow$ Mean $>$ Median

Left skewness $\Rightarrow$ Mean $<$ Median

Application

Skewness is useful when deciding whether mean or median is a more appropriate measure of central tendency.

12.1 Skewness – Figure



The figure compares distributions with different directions of asymmetry.

A longer tail toward the left indicates negative skewness, while a longer tail toward the right indicates positive skewness.

A symmetric distribution has approximately equal tails on both sides.

12.2 Skewness – Example

Example: Income Distribution

Suppose most people in a dataset have moderate incomes, while a small number have very high incomes.

The distribution will have a long tail toward larger values.

Right-skewed distribution

Typically,

$$\text{Mean} > \text{Median}$$

because the few high-income observations pull the mean toward the right.

12.3 Skewness – Application

Application

Income, house prices, transaction amounts, and many financial variables can exhibit right-skewed distributions.

Why it matters in AI

Highly skewed features can affect statistical analysis and machine learning models.

Possible preprocessing techniques include:

- Log transformation
- Robust scaling
- Other suitable transformations

13. Kurtosis

Definition

Kurtosis describes the degree to which a distribution has **heavy or light tails** relative to a normal distribution.

It is related to the tendency of a distribution to produce extreme observations.

Types

- **Mesokurtic:** Similar tail behaviour to a normal distribution.
- **Leptokurtic:** Heavier tails and greater tendency for extreme observations.
- **Platykurtic:** Lighter tails and fewer extreme observations.

Application

Kurtosis is useful when analyzing distributions where the presence of extreme observations is important.

13.1 Kurtosis – Excess Kurtosis

For a random variable X with mean μ and standard deviation σ , kurtosis can be expressed as

$$K = \frac{E[(X - \mu)^4]}{\sigma^4}$$

The **excess kurtosis** is

$$K_{\text{excess}} = K - 3$$

For a normal distribution:

$$K = 3$$

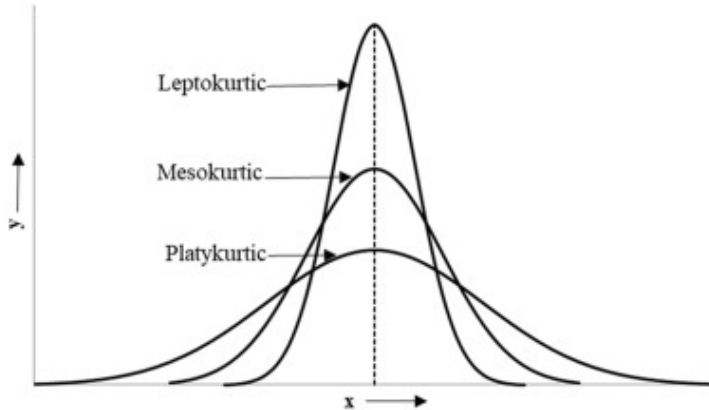
and therefore

$$K_{\text{excess}} = 0$$

Interpretation

Higher kurtosis generally indicates heavier tails and a greater tendency for extreme observations.

13.2 Kurtosis – Figure



The figure compares distributions with different tail behaviors.

Distributions with heavier tails have a greater tendency to produce observations far from the center.

13.3 Skewness vs Kurtosis

Measure	Describes
Skewness	Asymmetry of the distribution
Kurtosis	Tail heaviness / tendency toward extremes

Key distinction

- **Skewness** focuses on the direction and degree of asymmetry.
- **Kurtosis** focuses mainly on the behaviour of the tails and extreme observations.

Application in AI

Both measures can help identify unusual feature distributions and guide data preprocessing and exploratory analysis.

14. Population and Sample

Population

A population is the **complete set of individuals, objects, or observations** of interest in a study.

Sample

A sample is a **subset of the population** selected for analysis.

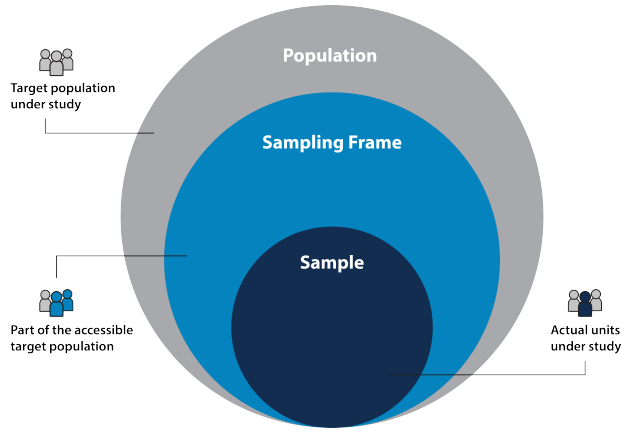
$$\text{Population} \supseteq \text{Sample}$$

Example

If a university has 5,000 students and we select 200 students for analysis:

- Population = 5,000 students
- Sample = 200 selected students

14.1 Population and Sample – Figure



The figure illustrates the relationship between a population, a sampling frame, and a sample in a research study.

The large gray circle represents the population, the entire target population under study.

The blue circle inside it represents the sampling frame, the portion of the target population from which participants can be selected.

The dark navy circle represents the sample, the actual units or individuals selected for the study.

14.2 Population and Sample – Comparison

	Population	Sample
Size	Complete set	Subset
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard deviation	σ	s
Purpose	Describe population	Estimate population

Key idea

Since studying an entire population may be expensive or impractical, statistical analysis often uses a sample to learn about the population.

Application

Sampling is fundamental in machine learning, surveys, experiments and data analysis.

15. Sample Mean

Definition

The sample mean is the arithmetic average of the observations in a sample.
For a sample

$$x_1, x_2, \dots, x_n$$

the sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Interpretation

The sample mean provides an estimate of the population mean μ .

$$\bar{x} \approx \mu$$

The quality of this estimate generally improves as the sample size increases.

15.1 Sample Mean – Example

Example

Consider a sample of five observations:

$$X = \{12, 15, 18, 20, 25\}$$

The sample mean is

$$\bar{x} = \frac{12 + 15 + 18 + 20 + 25}{5}$$

$$\boxed{\bar{x} = 18}$$

Interpretation

The value 18 is the average of the observations in the sample and can be used as an estimate of the population mean.

16. Sample Variance

Definition

Sample variance measures the variability of observations around the sample mean.

For a sample $x_1, x_2, \dots, x_n$:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Why $n - 1$?

The denominator $n - 1$ provides an unbiased estimator of the population variance under the usual random-sampling assumptions.

Key idea

Sample variance measures how much the observations vary around their sample mean.

16.1 Sample Variance – Example

Example

Consider:

$$X = \{2, 4, 6\}$$

The sample mean is

$$\bar{x} = 4$$

Squared deviations:

$$(2 - 4)^2 = 4, \quad (4 - 4)^2 = 0, \quad (6 - 4)^2 = 4$$

$$s^2 = \frac{4 + 0 + 4}{3 - 1}$$

$$\boxed{s^2 = 4}$$

Interpretation

The sample variance quantifies the spread of the observations around their sample mean.

17. Order Statistics

Definition

Order statistics are the observations arranged in **ascending order**.

Given observations

$$x_1, x_2, \dots, x_n,$$

their ordered values are written as

$$x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$$

where $x_{(i)}$ denotes the i -th smallest observation.

Examples of order statistics

- $x_{(1)}$ – minimum
- $x_{(n)}$ – maximum
- Middle order statistic – median
- Quartiles – divide ordered data into parts

17.1 Order Statistics – Example

Example

Consider:

$$X = \{40, 10, 30, 50, 20\}$$

Arrange the observations in ascending order:

$$10, 20, 30, 40, 50$$

Therefore,

$$x_{(1)} = 10, \quad x_{(2)} = 20, \quad x_{(3)} = 30, \quad x_{(4)} = 40, \quad x_{(5)} = 50$$

Hence,

$$x_{(1)} = \min(X)$$

$$x_{(5)} = \max(X)$$

The middle observation $x_{(3)} = 30$ is the median.

18. Covariance

Definition

Covariance measures the **direction of the linear relationship** between two variables.

For random variables X and Y :

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Interpretation

- Positive covariance: variables tend to increase together.
- Negative covariance: one tends to increase when the other decreases.
- Covariance near zero: weak linear association.

Note

The magnitude of covariance depends on the units of the variables.

18.1 Covariance – Example

Example

Consider two variables:

$$X = \{1, 2, 3\}$$

$$Y = \{2, 4, 6\}$$

As X increases, Y also increases. Their means are:

$$\bar{x} = 2, \quad \bar{y} = 4$$

The deviations have the same sign:

$$(-1)(-2), \quad (0)(0), \quad (1)(2)$$

All products are non-negative. Therefore, the covariance is **positive**.

Interpretation

The variables show a positive linear relationship.

19. Sample Covariance

Definition

Sample covariance measures the linear relationship between two variables using sample observations.

For paired observations

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n),$$

the sample covariance is

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Interpretation

- $s_{XY} > 0$: positive association.
- $s_{XY} < 0$: negative association.
- $s_{XY} \approx 0$: weak linear association.

19.1 Sample Covariance – Example

Example

Consider two variables:

$$X = \{1, 2, 3\}, \quad Y = \{2, 4, 6\}$$

The sample means are:

$$\bar{x} = 2, \quad \bar{y} = 4$$

x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	Product
1	2	-1	-2	2
2	4	0	0	0
3	6	1	2	2

$$s_{XY} = \frac{2 + 0 + 2}{3 - 1} = \boxed{2}$$

Since $s_{XY} > 0$, X and Y have a positive covariance and tend to increase together.

20. Correlation

Definition

Correlation is a standardized measure of the **strength and direction of linear association** between two variables.

The population correlation coefficient is

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

For a sample, the corresponding measure is the **sample correlation coefficient** r .

$$-1 \leq r \leq 1$$

Interpretation

- $r = 1$: perfect positive linear relationship.
- $r = -1$: perfect negative linear relationship.
- $r \approx 0$: weak or no linear relationship.

20.1 Correlation – Relation to Covariance

The sample correlation coefficient can be written as

$$r = \frac{s_{XY}}{s_X s_Y}$$

where:

- s_{XY} = sample covariance
- s_X = sample standard deviation of X
- s_Y = sample standard deviation of Y

Key difference

- Covariance depends on the units of measurement.
- Correlation is standardized and always lies between -1 and 1 .

Application

Correlation is commonly used for feature analysis and understanding relationships between variables in datasets.

21. Sample Covariance Matrix

Definition

For a dataset containing multiple variables, the sample covariance matrix summarizes the covariance between every pair of variables.

For p variables:

$$S = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1p} \\ s_{21} & s_{22} & \cdots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{p1} & s_{p2} & \cdots & s_{pp} \end{bmatrix}$$

where

$$s_{ij} = \frac{1}{n-1} \sum_{k=1}^n (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

The diagonal elements represent the **sample variances** of the variables.

21.1 Sample Covariance Matrix – Example

Suppose a dataset contains three variables:

{Age, Study Hours, Marks}

A sample covariance matrix may be represented as

$$S = \begin{bmatrix} s_{\text{Age, Age}} & s_{\text{Age, Study}} & s_{\text{Age, Marks}} \\ s_{\text{Study, Age}} & s_{\text{Study, Study}} & s_{\text{Study, Marks}} \\ s_{\text{Marks, Age}} & s_{\text{Marks, Study}} & s_{\text{Marks, Marks}} \end{bmatrix}$$

Interpretation

- Diagonal entries represent variances.
- Off-diagonal entries represent covariances.
- The matrix is symmetric:

$$s_{ij} = s_{ji}$$

21.2 Sample Covariance Matrix – Application

Why it is useful

The covariance matrix provides a compact representation of relationships among multiple numerical variables.

Applications in AI and Computing

- Feature analysis
- Principal Component Analysis (PCA)
- Multivariate statistical analysis
- Understanding relationships between features

Key idea

Multiple variables $\longrightarrow$ Covariance Matrix $\longrightarrow$ Multivariate Analysis

22. Frequentist Statistics

Definition

Frequentist statistics is an approach in which unknown population parameters are studied using data obtained from repeated sampling.

Key ideas

- Parameters are fixed but unknown quantities.
- Samples are considered random.
- Statistical conclusions are based on observed data.
- Sampling distributions are used to study the behavior of statistical estimators.

Example

A population has an unknown mean μ . A sample is collected and its mean $\bar{x}$ is used to estimate μ .

23. Sampling

Definition

Sampling is the process of selecting a subset of observations from a population for statistical analysis.

Why sampling?

- Studying the entire population may be expensive.
- It may be time-consuming or impractical.
- A representative sample can provide useful information about the population.

Simple Random Sampling

Each member of the population has an equal probability of being selected.

Population $\longrightarrow$ Sample $\longrightarrow$ Statistical Analysis

23.1 Sampling – Example

Example

Suppose a university has

$$N = 5000$$

students and we want to estimate the average number of hours students spend studying each day. Instead of collecting information from all 5,000 students, we select a random sample of $n=200$ students.

The sample mean

$$\bar{x}$$

can then be used to estimate the population mean

$$\mu.$$

The sample should be selected so that it reasonably represents the population.

24. Sampling Distribution

Definition

A sampling distribution is the probability distribution of a statistic obtained from all possible samples of a given size from a population.

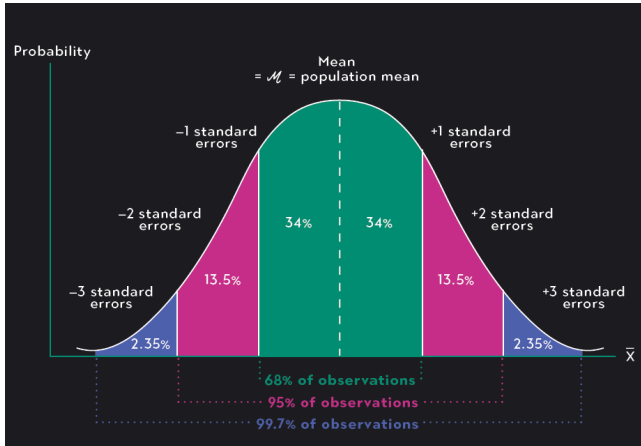
For example, the sample mean $\bar{X}$ has its own sampling distribution.

Population $\rightarrow$ Repeated Samples $\rightarrow$ Statistic $\rightarrow$ Sampling Distribution

Important

A sampling distribution describes the **variability of a statistic across repeated samples**, not the distribution of the individual observations.

24.1 Sampling Distribution – Figure



The bell curve is centered at the population mean μ .

Data are distributed into standard-error ranges around the mean.

About 68% of observations fall within ± 1 standard error.

About 95% fall within ± 2 standard errors.

About 99.7% fall within ± 3 standard errors.

The shaded sections represent the percentage of observations in each range.

24.2 Sampling Distribution – Sample Mean

Suppose the population has mean μ and variance σ^2 . For random samples of size n , the sample mean is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Its sampling distribution has:

$$E[\bar{X}] = \mu$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

Therefore,

$$\text{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

This quantity is called the **standard error** of the sample mean.

25. Standard Error

Definition

Standard error (SE) measures the variability of a statistical estimator across repeated samples.

For the sample mean:

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

When the population standard deviation σ is unknown, it is commonly estimated using the sample standard deviation s :

$$SE(\bar{X}) = \frac{s}{\sqrt{n}}$$

Key idea

As the sample size increases, the standard error of the sample mean decreases.

25.1 Standard Error – Example

Example

Suppose a sample has:

$$s = 10, \quad n = 100$$

The standard error of the sample mean is

$$\begin{aligned} SE(\bar{X}) &= \frac{s}{\sqrt{n}} \\ &= \frac{10}{\sqrt{100}} = \frac{10}{10} \\ &\boxed{SE(\bar{X}) = 1} \end{aligned}$$

Interpretation

A larger sample size generally produces a smaller standard error, making the sample mean a more precise estimator of the population mean.

26. Estimator, Estimate and Parameter

Parameter

A parameter is a fixed but unknown numerical characteristic of a population.

$$\mu, \quad \sigma^2, \quad p$$

Estimator

An estimator is a rule or statistic calculated from sample data that is used to estimate a population parameter.

$$\hat{\theta}$$

Estimate

An estimate is the numerical value obtained after applying the estimator to a particular sample.

Sample Data $\longrightarrow$ Estimator $\longrightarrow$ Parameter

26.1 Estimator, Estimate and Parameter – Example

Suppose μ is the unknown population mean.

The sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

is an **estimator** of μ .

For a particular sample:

$$X = \{10, 20, 30, 40, 50\}$$

we obtain

$$\bar{x} = 30.$$

Therefore:

- μ = population parameter
- $\bar{X}$ = estimator
- 30 = estimate

27. Bias and Variance

Bias

Bias measures the difference between the expected value of an estimator and the true population parameter.

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

An estimator is **unbiased** if

$$E[\hat{\theta}] = \theta$$

Variance

Variance measures how much an estimator varies across different samples.

$$\text{Var}(\hat{\theta}) = E[(\hat{\theta} - E[\hat{\theta}])^2]$$

Key idea

Bias describes systematic error, while variance describes sampling variability.

27.1 Bias and Variance – Trade-off

An estimator may have:

- Low bias but high variance.
- High bias but low variance.
- Both low bias and low variance.
- Both high bias and high variance.

Bias–Variance Trade-off

In statistical modelling and machine learning, reducing bias may increase variance and vice versa.

Good estimation $\approx$ Balance between Bias and Variance

Application

The bias–variance trade-off is important when selecting and evaluating machine learning models.

28. Mean Square Error

Definition

Mean Square Error (MSE) measures the average squared difference between an estimator and the true parameter.

For an estimator $\hat{\theta}$ of parameter θ :

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

MSE can be decomposed into:

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$$

Key idea

MSE combines both the variability and systematic error of an estimator.

28.1 Mean Square Error – Interpretation

The decomposition

$$MSE = \text{Variance} + \text{Bias}^2$$

shows that estimation error can arise from two sources.

- **Variance:** Estimator changes across different samples.
- **Bias:** Estimator systematically differs from the true parameter.

Goal

A good estimator should generally have a small MSE.

Application

MSE is widely used for evaluating statistical estimators and regression models.

29. Consistency

Definition

An estimator is consistent if it approaches the true population parameter as the sample size becomes very large.

For an estimator $\hat{\theta}_n$ of θ :

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as } n \rightarrow \infty$$

where $\xrightarrow{P}$ denotes convergence in probability.

Interpretation

As more data becomes available, a consistent estimator becomes increasingly close to the true parameter.

Example

The sample mean $\bar{X}$ is a consistent estimator of the population mean μ under standard random-sampling assumptions.

30. Central Limit Theorem

Definition

The Central Limit Theorem (CLT) states that, under suitable conditions, the sampling distribution of the sample mean approaches a normal distribution as the sample size becomes large.

If the population has mean μ and finite variance σ^2 :

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

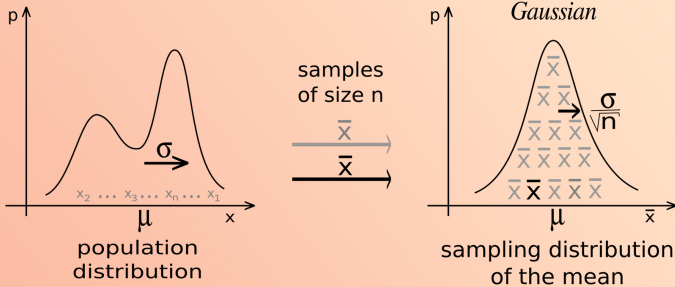
where $\xrightarrow{d}$ denotes convergence in distribution.

Key idea

The original population distribution does not have to be normal for the CLT to apply, provided suitable conditions hold.

30.1 Central Limit Theorem – Figure

Central Limit Theorem (CLT)



Regardless of the original population's shape, the distribution of sample means becomes approximately Gaussian (normal) as sample size n increases.

It is centered at the population mean μ , with standard error $\frac{\sigma}{\sqrt{n}}$, showing that larger samples produce more precise estimates.

30.2 Central Limit Theorem – Application

Why is the CLT important?

The CLT provides the theoretical basis for many statistical procedures involving sample means.

It is used in:

- Confidence interval construction
- Hypothesis testing
- Approximation of sampling distributions
- Statistical inference from large samples

$$n \uparrow \Rightarrow SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} \downarrow$$

Larger samples generally produce more stable estimates of the population mean.

31. Confidence Intervals

Definition

A confidence interval is an interval calculated from sample data that provides a range of plausible values for an unknown population parameter.

For a population mean with known σ , a confidence interval can be written as:

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where $z_{\alpha/2}$ is the appropriate standard normal critical value.

General form

$$\text{Estimate} \pm \text{Margin of Error}$$

31.1 Confidence Intervals – Example

Example

Suppose:

$$\bar{x} = 50, \quad \sigma = 10, \quad n = 100$$

For a 95% confidence interval:

$$z_{0.025} = 1.96$$

The margin of error is

$$1.96 \frac{10}{\sqrt{100}} = 1.96$$

Therefore,

$$CI = 50 \pm 1.96$$

$$CI = (48.04, 51.96)$$

Using this procedure, the interval obtained from the sample is intended to capture the true population mean in the long-run at the stated confidence level.

31.2 Confidence Intervals – Key Points

The width of a confidence interval depends on:

- Sample size
- Variability of the data
- Chosen confidence level

For the mean:

$$\text{Margin of Error} = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Effect of sample size

$$n \uparrow \Rightarrow \frac{\sigma}{\sqrt{n}} \downarrow$$

Therefore, larger samples generally lead to **narrower confidence intervals**.

32. Parametric Model Estimation

Definition

Parametric model estimation assumes that the population distribution belongs to a known family of distributions described by a finite number of parameters.

Examples

- Normal distribution – parameters μ and σ^2
- Bernoulli distribution – parameter p
- Poisson distribution – parameter λ

A parametric model can be written as

$$f(x; \theta)$$

where θ represents the unknown parameter(s).

Goal

Estimate the unknown parameters using observed sample data.

32.1 Parametric Model Estimation – Example

Example

Suppose a population is assumed to follow a normal distribution:

$$X \sim N(\mu, \sigma^2)$$

The parameters are:

μ = population mean

σ^2 = population variance

Given a sample, we can estimate these parameters using:

$$\hat{\mu} = \bar{x}$$

$$\hat{\sigma}^2 = s^2$$

The assumed distribution provides a mathematical model, while the sample data is used to estimate its unknown parameters.

33. Maximum Likelihood Estimation

Definition

Maximum Likelihood Estimation (MLE) is a method for estimating unknown parameters by choosing the parameter values that make the observed data most likely.

For observations

$$x_1, x_2, \dots, x_n$$

the likelihood function is

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta)$$

The maximum likelihood estimator is

$$\hat{\theta} = \arg \max_{\theta} L(\theta)$$

Choose the parameter value that gives the highest likelihood for the observed data.

33.1 Maximum Likelihood Estimation – Log-Likelihood

Since likelihoods involve products, it is often easier to maximize the logarithm of the likelihood.

The log-likelihood is

$$\ell(\theta) = \log L(\theta)$$

Therefore,

$$\hat{\theta} = \arg \max_{\theta} \ell(\theta)$$

Because the logarithm is a monotonically increasing function,

$$\arg \max_{\theta} L(\theta) = \arg \max_{\theta} \log L(\theta)$$

Advantage

The logarithm converts products into sums, making mathematical calculations easier.

33.2 Maximum Likelihood Estimation – Example

Example: Bernoulli Distribution

Let

$$X \in \{0, 1\}, \quad P(X = x) = p^x(1 - p)^{1-x}$$

where p is the unknown probability of success.

For observations $x_1, \dots, x_n$, the likelihood is

$$L(p) = \prod_{i=1}^n p^{x_i}(1 - p)^{1-x_i}$$

The MLE is obtained by maximizing $L(p)$:

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n x_i$$

$\hat{p}$ is the **sample proportion of successes**.

34. Non-parametric Model Estimation

Definition

Non-parametric model estimation does not assume that the population distribution belongs to a specific parametric family.

Instead, the distribution is estimated directly from the observed data.

Examples

- Empirical Cumulative Distribution Function (ECDF)
- Kernel Density Estimation (KDE)

Key idea

Data $\longrightarrow$ Estimate Distribution

without requiring a fixed distribution such as normal, Poisson, or Bernoulli.

Advantage

Flexible when the underlying distribution is unknown or does not fit a simple parametric family.

35. Empirical CDF

Definition

The Empirical Cumulative Distribution Function (ECDF) estimates the cumulative distribution function directly from sample data.

For a sample

$$x_1, x_2, \dots, x_n,$$

the empirical CDF is

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(x_i \leq x)$$

where $\mathbf{1}(\cdot)$ is the indicator function.

Interpretation

$F_n(x)$ represents the proportion of sample observations that are less than or equal to x .

35.1 Empirical CDF – Example

Consider the ordered sample:

$$X = \{10, 20, 30, 40, 50\}$$

Since $n = 5$:

$$F_n(x) = \frac{\text{Number of observations} \leq x}{5}$$

$$F_n(30) = 0.6$$

This means that 60% of the observations are less than or equal to 30.

x	$F_n(x)$
10	$1/5 = 0.2$
20	$2/5 = 0.4$
30	$3/5 = 0.6$
40	$4/5 = 0.8$
50	$5/5 = 1.0$

36. Kernel Density Estimation

Definition

Kernel Density Estimation (KDE) is a non-parametric method for estimating the probability density function of a dataset.

For observations $x_1, \dots, x_n$:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$

where:

- K = kernel function
- h = bandwidth
- n = number of observations

36.1 Kernel Density Estimation – Bandwidth

The **bandwidth** h controls the smoothness of the estimated density.

Small bandwidth

$h \downarrow \Rightarrow$ More detailed / less smooth estimate

Large bandwidth

$h \uparrow \Rightarrow$ Smoother estimate

Key idea

The bandwidth determines how much neighboring observations influence the estimated density.

Application

KDE is useful for visualizing and estimating unknown continuous distributions.

36.2 Kernel Density Estimation – Example

Example

Consider the sample:

$$X = \{2, 3, 3, 4, 5\}$$

KDE places a small smooth kernel around each observation.

The individual kernels are combined to produce an estimated density:

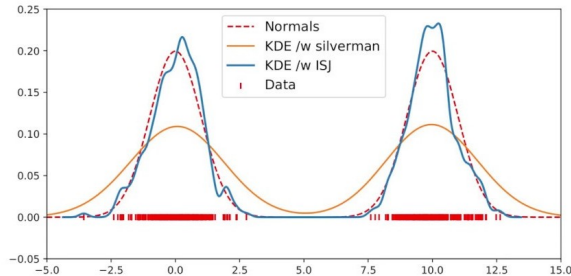
Individual Kernels $\longrightarrow$ Combined Density Estimate

Interpretation

Instead of assuming that the data follows a normal distribution, KDE estimates the shape of the distribution directly from the data.

36.3 Kernel Density Estimation – Figure

The *Improved Sheather Jones* (ISJ) algorithm is more robust with respect to *multimodality*.



The figure shows a smooth density curve estimated from observed data points.

Unlike a histogram, KDE produces a continuous estimate of the underlying distribution.

37. Parametric vs Non-parametric Model Estimation

Feature	Parametric	Non-parametric
Distribution assumption	Specific family	No fixed family
Parameters	Finite number	More flexible
Flexibility	Lower	Higher
Examples	Normal, Poisson	ECDF, KDE
Data requirement	Often fewer observations	Often more data needed

Key idea

Parametric → Assume a distribution

Non-parametric → Estimate from data

37.1 Parametric vs Non-parametric – Application

Parametric estimation

Useful when a suitable probability distribution can reasonably describe the data.

Non-parametric estimation

Useful when the underlying distribution is unknown, irregular, or difficult to represent using a simple parametric model.

In AI and Data Science

The choice depends on:

- Amount of available data
- Distributional assumptions
- Required flexibility
- Computational cost
- Purpose of the analysis

38. Applications in Computing and AI

Descriptive statistics provides a foundation for understanding and analyzing data in computing and Artificial Intelligence.

- **Data Analysis**

- Summarizes large datasets using mean, median, variance and other measures.

- **Data Preprocessing**

- Histograms and box plots help identify distributions and potential outliers.

- **Feature Analysis**

- Covariance and correlation help study relationships between variables.

- **Machine Learning**

- Statistical concepts support model estimation, evaluation and uncertainty analysis.

- **AI Applications**

- Used in areas such as prediction, classification, anomaly detection and pattern analysis.

39. Conclusion

- Descriptive statistics provides systematic methods for summarizing and understanding data.
- Measures of central tendency describe the typical value of a dataset.
- Measures of dispersion describe the variability of data.
- Graphical methods such as histograms and box plots provide visual understanding of distributions and outliers.
- Covariance and correlation help identify relationships between variables.
- Frequentist concepts provide a foundation for sampling, estimation and statistical inference.
- Parametric and non-parametric methods provide different approaches for estimating underlying data distributions.
- These concepts form an important mathematical foundation for data science, machine learning and Artificial Intelligence.

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Thank You

Arya T S